

## A UAV-based high-resolution vehicle trajectory dataset for motorcycle-dominated traffic

Data Description v1.2 (last updated 02 June 2026)

Table 1. Characteristics of the study sites in the AMIT-SI module.

| Intersection | Location Name                                   | City                                   | Geometry              | Traffic control                                              | Time of survey | Total Duration | Vehicle types (> 1%)           |
|--------------|-------------------------------------------------|----------------------------------------|-----------------------|--------------------------------------------------------------|----------------|----------------|--------------------------------|
| A01          | Zhongxiao Rd and Zhongzheng Rd                  | Xinhua Dist., Tainan City, Taiwan      | Four-leg (orthogonal) | Signalized, two-phase                                        | Evening peak   | 77.5 min       | Passenger car, PTW, truck      |
| A02          | Sec. 1, Beiyi Rd. & Wenzhong Rd.,               | Xindian Dist., New Taipei City, Taiwan | Four-leg (orthogonal) | Signalized, three-phase with LPI                             | Morning peak   | 66.1 min       | Passenger car, PTW, truck, bus |
| A03          | Xinzhong N. Rd. & Rongmin Rd. & Rongmin S. Rd., | Zhongli Dist., Taoyuan City, Taiwan    | Four-leg (skewed)     | Signalized, two-phase                                        | Morning peak   | 39.6 min       | Passenger car, PTW, truck      |
| A04          | Jingwu Rd. & Bo'ai Rd.                          | Fengshan Dist., Kaohsiung City, Taiwan | Four-leg (skewed)     | Signalized, three-phase with leading phase                   | Evening peak   | 46.1 min       | Passenger car, PTW, truck      |
| A05          | Minguang E. Rd. & Chunri Rd.                    | Taoyuan Dist., Taoyuan City, Taiwan    | Four-leg (orthogonal) | Signalized, four-phase with leading phase and lagging phases | Evening peak   | 82.2 min       | Passenger car, PTW, truck      |
| A06          | Zhonghua Rd. & Fuxing Rd. & Guangming 11th Rd.  | Zhubei City, Hsinchu County, Taiwan    | Four-leg (orthogonal) | Signalized, four-phase                                       | Morning peak   | 80.1 min       | Passenger car, PTW, truck      |

Note:

PTW = powered-two-wheeler, including scooters, motorcycles, and other motorized two-wheelers.

LPI = leading pedestrian interval

Table 2. Characteristics of the study sites in the AMIT-BA module.

| Intersection | Location Name                                                  | City                                     | Geometry               | Traffic control                                                                       | Main engineering improvement                                | Time of survey | Total Duration | Vehicle types (> 1%)    |
|--------------|----------------------------------------------------------------|------------------------------------------|------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------|----------------|----------------|-------------------------|
| B01          | Zhongzheng Rd and Xintai Rd. (Before engineering improvement)  | Xinzhuang Dist., New Taipei City, Taiwan | Four-leg (orthogonal)  | Signalized, five-phase with exclusive pedestrian phase and leading and lagging phases | Added auxiliary left-turn lane and signal timing adjustment | Morning peak   | 85.7 min       | Passenger car, PTW, bus |
| B02          | Zhongzheng Rd and Xintai Rd. (After engineering improvement)   | Xinzhuang Dist., New Taipei City, Taiwan | Four-leg (orthogonal)  | Signalized, five-phase with exclusive pedestrian phase and leading and lagging phases |                                                             | Morning peak   | 84.8 min       | Passenger car, PTW, bus |
| B03          | Fuxing Rd. and Minsheng Rd. (Before engineering improvement)   | Taoyuan Dist., Taoyuan City, Taiwan      | Four-leg (skewed)      | Signalized, three-phase with leading phase                                            | Left-turn guidance markings and channelization improvements | Mid-morning    | 87.0 min       | Passenger car, PTW, bus |
| B04          | Fuxing Rd. and Minsheng Rd. (After engineering improvement)    | Taoyuan Dist., Taoyuan City, Taiwan      | Four-leg (skewed)      | Signalized, three-phase with leading phase                                            |                                                             | Mid-morning    | 47.6 min       | Passenger car, PTW, bus |
| B05          | Zhonghua Rd and Chenggong Rd. (Before engineering improvement) | Zhongli Dist., Taoyuan City, Taiwan      | Three-leg (orthogonal) | Signalized, three-phase with LPI                                                      | Dedicated PTW left-turn facility and signal phase           | Morning peak   | 77.1 min       | Passenger car, PTW, bus |
| B06          | Zhonghua Rd and Chenggong Rd. (After engineering improvement)  | Zhongli Dist., Taoyuan City, Taiwan      | Three-leg (orthogonal) | Signalized with dynamic timing, three-phase with LPI                                  |                                                             | Morning peak   | 88.7 min       | Passenger car, PTW, bus |

Note:

PTW = powered-two-wheeler, including scooters, motorcycles, and other motorized two-wheelers.

LPI = leading pedestrian interval